



INTEGRAL CALCULUS WORKSHOP

“PORTFOLIO”

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Introduction

This portfolio of evidence corresponds to the Integral Calculus course and has the purpose of compiling and presenting, in an organized manner, the theoretical notes, problem-solving procedures, and activities completed throughout the semester. Through this work, we aim to reflect the knowledge acquired as well as the development of skills to apply mathematical concepts in different contexts.

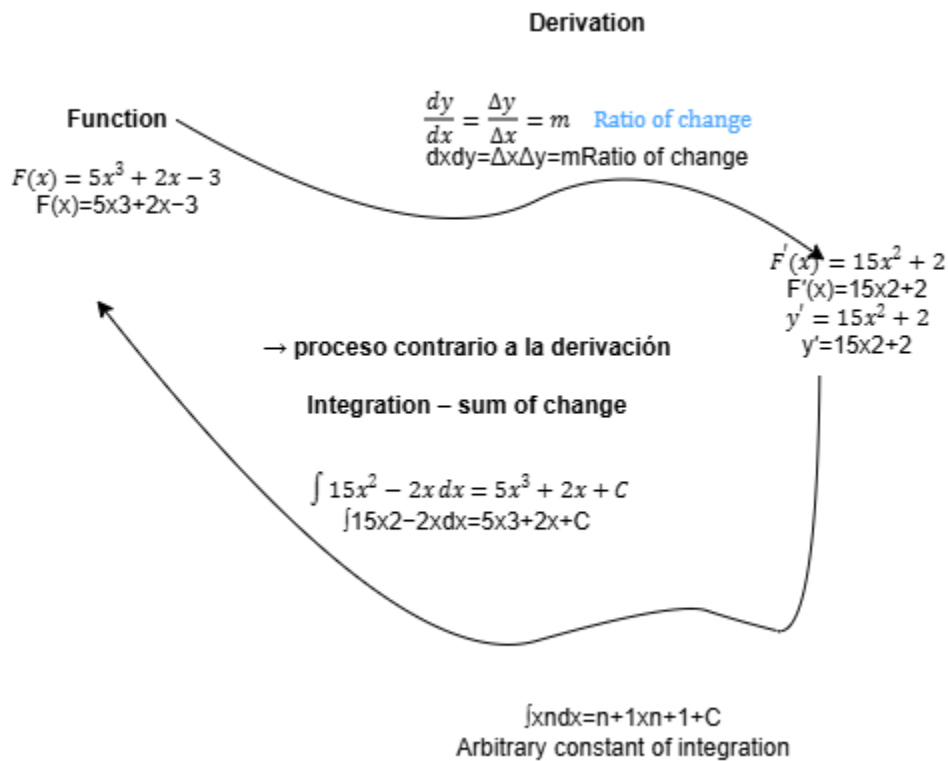
The study of Integral Calculus represents a fundamental tool within our academic formation, as it provides solid foundations for the analysis and resolution of problems in the field of exact sciences and in our area of study. During the course, we explored definitions, properties, rules, and theorems that allowed us to understand more deeply the foundations of calculus, while also strengthening our logical reasoning and critical thinking skills.

Furthermore, this portfolio seeks to highlight not only the academic progress achieved but also the perseverance and discipline applied when facing mathematical challenges. The organization of the material aims to demonstrate how each topic studied is interconnected, contributing to an integral vision of the subject and reinforcing competencies that will be useful in later stages of our university development.

The process of integration

Logic of a derivative: A rate of change.

The constant of a derivative is =0, because it does not change.



The general solution of integrals of the ax^n

a = Coefficient
 x = Independent variable
 n = grade, degree or power

This rule when n is fractional, zero, or positive or negative integer with the exception of $n = -1$

Demonstration

Problem 1. Determine

$$a) \int 5x^2 dx$$

$$b) \int 2t^3 dt$$

$$\int f(x) * dx = 5 * \frac{x^3}{3} + C$$

$$F(x) = \frac{1}{2} * t^4 + c$$

$$F(x) = \frac{5}{3} * x^3 + c$$

Problem 2. Determine

$$\int (4 + \frac{3}{7}x - 6x^2) dx$$

$$\int (4dx + \int \frac{3}{7}x dx - \int 6x^2 dx$$

$$4 \int dx + \frac{3}{7} \int x dx - 6 \int x^2 dx$$

$$4x + \frac{3}{7} (\frac{x^2}{2}) - 6 (\frac{x^3}{3}) + c$$

$$4x + \frac{3}{7} x^2 - 2x^3 + c$$

Problem 3. Determine a and b

$$(a) \int (2x^3 - 3x)/(4x) dx$$

$$\int (2x^3/4x - 3x/4x) dx$$

$$= \int (x^2/2 - 3/4) dx$$

$$1/2 \int x^2 dx - 3/4 \int dx$$

$$= 1/2 [x^3/3] - 3/4 [x] + c$$

$$F(x) = 1/6 x^3 - 3/4 x + c$$

$$(b) \int (1 - t)^2 dt$$

$$\int (1 - 2t + t^2) dt$$

$$= \int 1 dt - 2 \int t dt + \int t^2 dt$$

$$= [t] - 2[t^2/2] + [t^3/3] + c$$

$$= t - t^2 + 1/3 t^3 + c$$

Problem 4. Determine $\int 3/x^2 dx$

$$3 \int x^{-2} dx$$

$$3 [x^{-1}/(-1)] + c$$



$$= -3/x + c$$

Problem 5. Determine $\int 3\sqrt{x} \, dx$

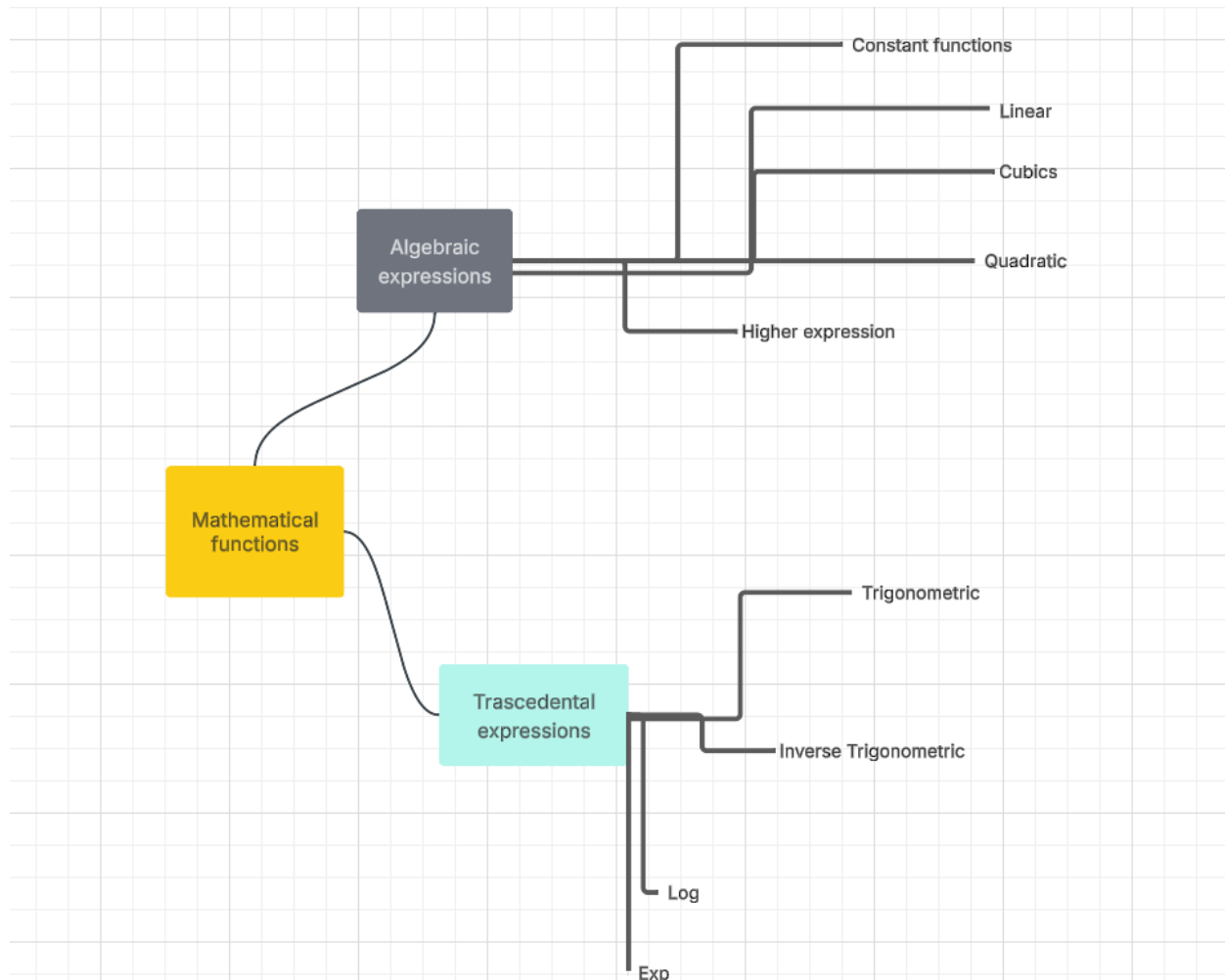
$$\begin{aligned} & \int 3x^{(1/2)} \, dx \\ & 3 \int x^{(1/2)} \, dx \\ & = 3 [x^{(3/2)}/(3/2)] + c \\ & = 3 [2/3 x^{(3/2)}] + c \\ & F(x) = 2 x^{(3/2)} + c \end{aligned}$$

Problem 6. Determine $\int \frac{-5}{9\sqrt[4]{t^3}} \, dt$

$$\begin{aligned} \int \frac{-5}{9\sqrt[4]{t^3}} \, dt &= \int \left(\frac{-5}{9} - \frac{t^{-\frac{3}{4}}}{9} \right) dt = \int \frac{-5}{9t^{\frac{3}{4}}} \, dt = -\frac{5}{9} \int t^{-\frac{3}{4}} \, dt = -\frac{5}{9} \left[\frac{t^{-\frac{1}{4}}}{-\frac{1}{4}} \right] + c = \\ & -\frac{5}{9} \left[4t^{\frac{1}{4}} \right] + c = -\frac{20}{9} t^{\frac{1}{4}} + c = -\frac{20}{9} \sqrt[4]{t} + c \end{aligned}$$

Problem 7. Determine $\int (1 + \theta)^2 / \sqrt{\theta} \, d\theta$

$$\begin{aligned} & \int (1 + 2\theta + \theta^2) / \theta^{(1/2)} \, d\theta \\ & \int \theta^{(-1/2)} \, d\theta + 2 \int \theta^{(1/2)} \, d\theta + \int \theta^{(3/2)} \, d\theta \\ & = [\theta^{(1/2)}/(1/2)] + 2[\theta^{(3/2)}/(3/2)] + [\theta^{(5/2)}/(5/2)] + c \\ & = 2\theta^{(1/2)} + 4/3 \theta^{(3/2)} + 2/5 \theta^{(5/2)} + c \end{aligned}$$



Problem 8. Determine a and b

$$(a) \int 4\cos(3x)dx$$

$$= 4/3 [\sin(3x)] + c$$

$$= (4/3)\sin(3x) + c$$

$$4\int \cos(3x)dx$$

$$u = 3x$$

$$du = 3dx$$

$$\int \cos(u)du = \sin(u)$$

$$4\int \cos(3x) \cdot 3dx = 4\int \cos(3x)dx$$

$$(b) \int 5\sin(2\theta)d\theta$$

$$5\int \sin(2\theta)d\theta$$

$$u = 2\theta$$

$$\begin{aligned} du &= 2d\theta \\ 5 \int \sin(u) du & \\ &= 5/2 \int \sin(2\theta) \cdot 2d\theta \end{aligned}$$

$$\begin{aligned} &= 5/2 [-\cos(2\theta)] + c \\ &= -5/2 \cos(2\theta) + c \end{aligned}$$

Problem 9. Determine a and b

$$(a) \int \sec^2(4t) dt$$

$$F(t) = 1/4 \tan(4t) + c$$

$$\begin{aligned} \int \sec^2(4t) dt & \\ u &= 4t \\ du &= 4dt \\ 1/4 \int \sec^2(u) \cdot 4dt & \\ &= 1/4 [\tan(u)] + c \end{aligned}$$

$$\begin{aligned} (b) \int 3 \operatorname{cosec}^2(2\theta) d\theta & \\ 3 \int \operatorname{cosec}^2(2\theta) d\theta & \\ 1/2 \cdot 3 \int \operatorname{cosec}^2(2\theta) \cdot 2d\theta & \\ &= 3/2 [-\cot(2\theta)] + c \\ &= -3/2 \cot(2\theta) + c \end{aligned}$$

Problem 10. Determine a and b

$$(a) \int 5e^{(3x)} dx$$

$$(b) \int 2/3 e^{(-4t)} dt$$

$$\begin{aligned} 5 \int e^{(3x)} dx & \\ u &= 3x \\ du &= 3dx \\ 5/3 \int e^u \cdot 3dx & \\ &= 5/3 [e^{(3x)}] + c \\ &= 5/3 e^{(3x)} + c \end{aligned}$$

$$\begin{aligned} 2/3 \int e^{(-4t)} dt & \\ u &= -4t \\ du &= -4dt \\ -1/2 \cdot 2/3 \int e^u \cdot (-4dt) & \\ &= -2/12 \int e^{(-4t)} \cdot (-4dt) \\ &= -1/6 \int e^{(-4t)} \cdot (-4dt) \\ &= -1/6 [e^{(-4t)}] + c \\ F(t) &= -1/6 e^{(-4t)} + c \end{aligned}$$

Problem 11. Determine

$$\begin{aligned} a) \int \frac{3}{5x} dx &= \frac{3}{5} \int x \cdot \frac{1}{x} dx = \frac{3}{5} \int \frac{1}{x} dx = \int \frac{du}{u} = \ln|u| + c = \frac{3}{5} [\ln|x|] + c \\ &= \frac{3 \ln|x|}{5} + c \end{aligned}$$

$$b) \int \left(\frac{2m^2 + 1}{m} \right) dm = \int \frac{2m^2}{m} dm + \int \frac{1}{m} dm = \int 2m dm + \int \frac{1}{m} dm$$

$$2\left[\frac{m^2}{2}\right] + \ln|m| + c = m^2 + \ln|m| + c$$

$$1a) \int 4dx = 4 \int dx = 4x + c$$

$$b) \int 7x dx = \frac{7x^2}{2} + c$$

$$2a) \int \frac{2}{5}x^2 dx = \frac{2}{15}x^3 + c$$

$$b) \int \frac{5}{6}x^3 dx = \frac{5}{24}x^4 + c$$

$$3a) \int \left(\frac{3x^2 - 5x}{x}\right) dx = \frac{3x^2}{2} - 5x + c$$

$$4a) \int \frac{4}{3x^2} dx = \frac{4}{3} \int \frac{1}{x^2} dx = \frac{4}{3} \int x^{-2} dx = \frac{4}{3} \left[\frac{x^{-1}}{-1} \right] + c = \frac{x^{0+1}}{0+1} = \frac{x^1}{1} = x = -\frac{4}{3}x^{-1} + c - \frac{4}{3x} + c$$

$$4b) \int (2 + \phi)^2 d\phi = \int (4 + 4\phi + \phi^2) d\phi = \int (4 + 4\phi + \phi^2) d\phi$$

$$\int 4d\phi + \int 4\phi d\phi + \int \phi^2 d\phi = \int 4\phi + 4\left[\frac{\phi^2}{2}\right] + \frac{\phi^3}{3} + c = 4\phi + 2\phi^2 + \frac{\phi^3}{3} + c$$

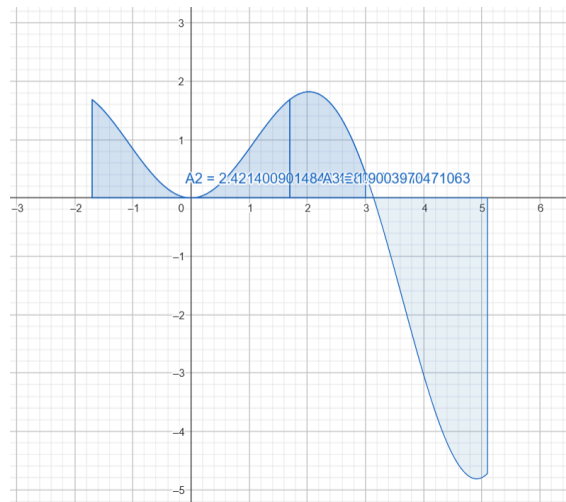
$$4b) \int \frac{3}{4x^4} dx = \frac{3}{12y^2} + c$$

The definite integrals

There are a number of applications of an integral calculus in engineering. The determination of areas mean and rms values, centroids and second moments of area and radius are included in this chapter.

Areas under and between curves

$$\text{Total area shaded area} = \int_a^b f(x)dx - \int_b^c f(x)dx + \int_c^a f(x)dx$$



Problem 1. Determine the area between the curve $y = x^3 - 2x^2 - 8x$ and the x-axis.

$$\begin{aligned} \text{Total area} &= \int_{-2}^0 (x^3 - 2x^2 - 8x) dx \\ A1 &= \left[\frac{x^4}{4} - \frac{2x^3}{3} - 4x^2 \right] \\ &= \left[\frac{0^4}{4} - \frac{2(0)^3}{3} - 4(0)^2 \right] \\ &= \left[\frac{(-2)^4}{4} - \frac{2(-2)^3}{3} - 4(-2)^2 \right] \\ &= 0 - \left[-\frac{20^3}{3} \right] \Rightarrow \frac{20}{3} \approx 6.67 \end{aligned}$$

$$A2 = \int_0^4 (x^3 - 2x^2 - 8x) dx$$

$$= \left[\frac{4^4}{4} - \frac{2(4)^3}{3} - 4(4)^2 \right] - 0$$

$$= -\frac{128}{3}u^2 \approx 42.6u^2$$

$$At = A1 - A2 \Rightarrow \frac{20}{3} - \left(-\frac{128}{3}\right) = \frac{148}{3}u^2 \approx 49.3u^2$$

Problem 2. Determine the area enclosed between the curves $y = x^2 + 1$ & $y = 7 - x$

$$\int_{-3}^2 (7-x)dx - \int_{-3}^2 (x^2 + 1)dx$$

$$= [7x - \frac{x^2}{2}]_{-3}^2 - [\frac{x^3}{3} + x]_{-3}^2$$

$$= [7(2) - \frac{2^2}{2}] - [7(-3) - \frac{(-3)^2}{2}] - [\frac{2^3}{3} + 2] - [-\frac{3^3}{3} - 3]$$

$$= [14 - 2] - [-21 - 9/2] - [8 + 2] - [-27 - 3]$$

$$= [12] - [-21 - 4.5] - [10] - [-30]$$

$$= [12] - [-25.5] - [10] - [-30]$$

$$= 12 + 25.5 - 10 + 30$$

$$= 57.5$$

$$\int_{-3}^2 (x^2 + 1)dx - \int_{-3}^2 (7-x)dx$$

$$= [\frac{x^3}{3} + x]_{-3}^2 - [7x - \frac{x^2}{2}]_{-3}^2$$

$$= [\frac{2^3}{3} + 2] - [-\frac{3^3}{3} - 3] - [7(2) - \frac{2^2}{2}] - [7(-3) - \frac{(-3)^2}{2}]$$

$$= [8 + 2] - [-27 - 3] - [14 - 2] - [-21 - 9/2]$$

$$= [10] - [-30] - [12] - [-25.5]$$

$$= [143] - [-12] \left[\frac{14}{3} \right] - [-12] [314] - [-12]$$

$$= [143] + 12 \left[\frac{14}{3} \right] + 12 [314] + 12$$

$$= 143 + 36 \frac{14}{3} + \frac{36}{3} 314 + 336$$

$$= 50/3 \approx 16.67 \frac{50}{3} \approx 16.6$$

$$\text{Area} = 37.5/2 - 50/3 \approx 18.75 - 16.67 \frac{37.5}{2} - \frac{50}{3} \approx 18.75 - 16.67$$

$$= 1256 \approx 20.83 \frac{125}{6} \approx 20.83 u^2 u^2$$

$$At = 1483 u^2 \approx 49.3 \frac{148}{3} u^2 \approx 49.3$$

Integration using algebraic substitutions

Functions which require integrating are not always in the "standard form". However, it is often possible to change a function into a form which can be integrated by using either:

- An algebraic substitution
- An trigonometric or hyperbolic substitution
- Partial fractions
- The $t = \tan \theta/2$ substitution
- Integration by parts
- Reduction formula

Algebraic substitutions

Problem 1. Determine $\int \cos((3x + 7))dx$

$$u = 3x + 7$$

$$du = 3dx$$

$$\frac{1}{3} \int \cos(u) \cdot 3dx$$

$$\int \cos(u) du = \sin(u)$$

$$\frac{1}{3} [\sin(3x + 7)] + c$$

$$F(x) = \sin(3x + 7)/3 + c$$

Problem 2. Find $\int (2x - 5)^7 dx$

$$u = 2x - 5$$

$$du = 2dx$$

$$\frac{1}{2} \int (2x - 5)^7 \cdot 2dx$$

$$\frac{1}{2} [(2x - 5)^8 / 8] + c$$

$$F(x) = \frac{1}{16} (2x - 5)^8 + c$$

Problem 3. Find $\int \frac{4}{5x - 3} dx$

$$u = 5x - 3$$

$$du = 5dx$$

$$\frac{4}{5} \int u^{-1} \cdot 5dx$$

$$\int (du/u) = \ln|u| + c$$

$$\frac{4}{5} [\ln|5x - 3|] + c$$

$$F(x) = \frac{4}{5} \ln|5x - 3| + c$$

Problem 4. Evaluate $\int_0^1 2e^{(6x - 1)} dx$, correct to 4 significant figures
 $\int_0^1 2e^{(6x - 1)} dx$

$$= 2 \int_0^1 e^{(6x - 1)} dx$$

$$u = 6x - 1$$

$$du = 6dx$$

$$\frac{1}{3} \int e^u \cdot 6dx$$

$$\int e^u du = e^u$$

$$\frac{1}{3} [e^{(6x - 1)}]_0^1$$

$$= \frac{1}{3} [e^{(6 \cdot 1 - 1)} - e^{(6 \cdot 0 - 1)}]$$

$$= \frac{1}{3} [e^5 - e^{-1}]$$

$$F(x) \approx 49.35 u^2$$

Integration Using Partial Fractions

Expressions which at first sight look impossible to integrate using standard techniques may in fact be integrated by first expressing them as simpler partial fractions and then using earlier learned techniques.

At the end of this chapter, you should be able to:

- Integrate functions using partial fractions with linear factors.
- Integrate functions using partial fractions with repeated linear factors.
- Integrate functions using partial fractions with quadratic factors.

Problem 1. Determine $\int (11 - 3x) / (x^2 + 2x - 3) dx$

1. Factorize the denominator:

$$\{11 - 3x\} / \{x^2 + 2x - 3\} = \{11 - 3x\} / \{(x + 3)(x - 1)\}$$

2. Separate into two simpler fractions:

$$\{11 - 3x\} / \{(x + 3)(x - 1)\} = \{A\} / \{x + 3\} + \{B\} / \{x - 1\}$$

3. Multiply the previous expression by the two obtained factors:

$$[(11 - 3x) / (x + 3)(x - 1)] = \{A\} / \{x + 3\} + \{B\} / \{x - 1\} [(x + 3)(x - 1)]$$

$$11 - 3x = A(x - 1) + B(x + 3)$$

4. Use a base of calculus to solve for A and B:

With $x = -3$

$$11 - 3(-3) = A(-3 - 1) + B(-3 + 3)$$

$$11 + 9 = -4A$$

$$20 = -4A$$

$$A = -5$$

With $x = 1$:

$$11 - 3(1) = A(1 - 1) + B(1 + 3)$$

$$11 - 3 = 4B$$

$$8 = 4B$$

$$B = 2$$

5. Replace the values for A and B in the new partial fractions:

$$\int \frac{11 - 3x}{x^2 + 2x - 3} dx = \int \frac{-5}{x + 3} dx + \int \frac{2}{x - 1} dx$$

$$u = x + 3$$

$$du = dx$$

$$u = x - 1$$

$$du = dx$$

$$= -5 \int \frac{dx}{x+3} + 2 \int \frac{dx}{x-1}$$

$$\int \frac{du}{u} = \ln |u| + C$$

$$F(x) = -5 \ln |x + 3| + 2 \ln |x - 1| + C$$

6. Apply the corresponding properties for log:

$$\log N^R = r \log N$$

$$F(x) = \ln |(x + 3)^{-5} + \ln (x - 1)^2 + C$$

$$\log M * N = \log M + \log N$$

$$F(x) = \ln [(x + 3)^{-5} * (x - 1)^2] + C$$

Problem 2. Find $\int \frac{2x^2 - 9x - 35}{[(x + 1)(x - 2)(x + 3)]} dx$

$$\left\{ \frac{2x^2 - 9x - 35}{(x + 1)(x - 2)(x + 3)} \right\} = \frac{A}{x + 1} + \frac{B}{x - 2} + \frac{C}{x + 3}$$

$$2x^2 - 9x - 35 = A(x - 2)(x + 3) + B(x + 1)(x + 3) + C(x + 1)(x - 2)$$

With $x = -1$:

$$2(-1)^2 - 9(-1) - 35 = A(-1 - 2)(-1 + 3)$$

$$2 + 9 - 35 = -3A$$

$$-24 = -3A$$

$$A = 8$$

With $x = 2$:

$$2(2)^2 - 9(2) - 35 = B(2 + 1)(2 + 3)$$

$$8 - 18 - 35 = 15B$$

$$-45 = 15B$$

$$B = -3$$

With $x = -3$:

$$2(-3)^2 - 9(-3) - 35 = C(-3 + 1)(-3 - 2)$$

$$18 + 27 - 35 = -10C$$

$$10 = -10C$$

$$C = -1$$

$$= 4 \int \frac{4}{x+1} dx + \int \frac{-3}{x-2} dx + \int \frac{-1}{x+3} dx$$

$$u = x + 1$$

$$du = dx$$

$$u = x - 2$$

$$du = dx$$

$$u = x + 3$$

$$du = dx$$

$$= 4 \ln(x+1) - 3 \ln(x-2) - \ln(x+3) + C$$

$$n \log M * N = \log M^n$$

$$= \ln(x+1)^4 + \ln(x-2)^{-3} + \ln(x+3)^{-1} + C$$

$$\log M * N = \log M + \log N$$

$$F(x) = \ln [(x+1)^4 \ln (x-2)^{-3} \ln (x+3)] + C$$

$$F(x) = \ln \left[\frac{(x+1)^4}{(x-2)^3 (x+3)} \right] + C$$

Problem 3 Determine

Problem 3 $\int \frac{x^2+1}{x^2+3x+2} dx$

$$= \int \left[1 + \frac{3x-1}{x^2-3x+2} \right] dx = x^2 - 3x + 2/x^2 + 1 = 1$$

$$= \left[\frac{3x-1}{x^2-3x+2} = \frac{A}{x+2} + \frac{B}{x-1} \right] = [(x-2)(x-1)]$$

$$= 3x - 1 = A(x-1) + B(x-2)$$

$$= 3(2) - 1 = A(2-1) + B(2-2)$$

$$= 6 - 1 = A1 + B0 \Rightarrow A = 5$$

$$= 3(1) - 1 = A(1-1) + B(1+2)$$

$$= 3 - 1 = A0 + B - 1 = 2 = -B \Rightarrow B = -2$$

$$\int 1 dx + 5 \int \frac{1}{x-2} dx - 2 \int \frac{1}{x-1} dx$$

$$= x + 5 \ln|x-2| - 2 \ln|x-1| + c$$

$$= x + \ln|x-2|^5 + \ln|x-1|^{-2} + c$$

$$F(x) = x + \ln \left[\frac{(x-2)^5}{(x-1)^2} \right] + c$$

Problem 4 $\int_2^3 \frac{x^3 + 2x^2 - 4x - 4}{x^2 + x - 2} dx$

$$= \int_2^3 \left(x - 3 + \frac{x+10}{(x+2)(x-1)} \right) dx$$

$$= \left[\frac{x-10}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1} \right] [(x+2)(x-1)]$$

$$= x - 10 = A(x+2) + B(x-1)$$

$$x = 1$$

$$1 - 10 = A(1+2) + B(1-1)$$

$$9 = A(3) + B(0)$$

$$9 = A(3)$$

$$\frac{9}{-3} = A$$

$$A = -3$$

$$x = -2$$

$$-2 - 10 = A(-2+2) + B(2-1)$$

$$-12 = A(0) + B(-3)$$

$$-12 = B(-3)$$

$$\frac{-12}{-3} = B$$

$$B = 4$$

$$= \int_2^3 \left(x - 3 - \frac{3}{x-1} + \frac{4}{x+2} \right) dx$$

$$= \left[\frac{x^2}{2} \right]_2^3 - 3[x]_2^3 - 3[\ln(x-1)]_2^3 + [\ln(x+2)]_2^3$$

$$= \left[\frac{3^2}{2} - \frac{2^2}{2} \right] - 3[3-2] - 3[\ln(3-1) - \ln(2-1)] + 4[\ln(3+2) - \ln(2+2)]$$

$$= \frac{5}{2} - 3 - 2.0794 + 0.88$$

$$F(x) = -1.699u^2$$

Problem 5 Determine $\int \frac{2x+3}{(x-2)^2} dx$

$$= \left[\frac{2x+3}{(x-2)^2} = \frac{A}{x-2} + \frac{B}{(x-2)^2} \right] [(x-2)^2]$$

$$= 2x + 3 = A(x-2) + B$$



$$x = 2$$

$$2(2) + 3 = B$$

$$B = 7$$

$$x = 1, B = 7$$

$$2(1) + 3 = A(1 - 2) + 7$$

$$5 = -A + 7$$

$$5 - 7 = -A$$

$$-2 = -A$$

$$A = 2$$

$$= \int \frac{2x+3}{(x-2)^2} dx = \int \frac{2}{x-2} dx + \int \frac{7}{(x-2)^2} dx$$

$$= \int 7(x-2)^{-2} dx = 7 \int (x-2)^{-2} dx = 7 \left[\frac{x-2}{-2} \right] + C$$

$$F(x) = 2 \ln(x-2) - \frac{7}{x-2} + C$$

Problem 6 Find $\int \frac{5x^2-2x-19}{(x+3)(x-1)^2} dx$

$$\frac{5x^2-2x-19}{(x+3)(x-1)^2} = \frac{A}{x+3} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$$

$$5x^2 - 2x - 19 = A(x-1)^2 + B(x+3)(x-1) + C(x+3)$$

$$x = -3$$

$$5(-3)^2 - 2(-3) - 19 = A(-3-1)^2$$

$$45 + 6 - 19 = A(16)$$

$$32 = A(16)$$

$$\frac{32}{16} = A$$

$$A = 2$$

$$x = 1$$

$$5(1)^2 - 2(1) - 19 = C(1+3)$$

$$5 - 2 - 19 = C(4)$$

$$-16 = C(4)$$

$$\frac{-16}{4} = C$$

$$C = -4$$

$$5(0)^2 - 2(0) - 19 = 2(0-1)^2 + B(0-1)(0+3) - 4(0+3)$$

$$-19 = 2 + B(-3)(-12)$$

$$-19 = -10 + B(-3)$$

$$-19 + 10 = B(-3)$$

$$\frac{9}{-3} = B$$

$$B = 3$$

$$\int \frac{5x^2 + 2x - 19}{(x+3)(x-1)^2} dx = \int \frac{2}{x+3} dx + \int \frac{3}{x-1} dx + \int \frac{-4}{(x-1)^2} dx$$

$$= 2 \int \frac{dx}{x+3} + 3 \int \frac{dx}{x-1} - 4 \int (x-1)^{-2} dx$$

$$= 2 \ln|x+3| + 3 \ln|x-1| - \frac{4(x-1)^{-1}}{-1} + C$$

$$F(x) = \ln[(x+3)^2(x-1)^3] + \frac{4}{x-1} + C$$

Problem 7 Evaluate $\int_{-2}^1 \frac{3x^2 + 16x + 15}{(x+3)^3} dx$

$$\int_{-2}^1 \frac{3x^2 + 16x + 15}{(x+3)^3} dx = u = x + 3 \quad du = dx$$

$$3x^2 + 16x + 15 = 3(u-3)^2 + 16(u-3) + 15 =$$

$$3(u^2 - 6u + 9) + 16u - 48 + 15 = 3u^2 - 18u + 27 + 16u - 48 + 15$$

$$3u^2 - 2u - 6$$

$$\int_1^4 \frac{3u^2 - 2u - 6}{u^3} du =$$

$$\int_1^4 (3u^{-1} - 2u^{-2} - 6u^{-3}) du = \left[3\ln|u| + \frac{2}{u} + \frac{3}{u^2} \right]_1^4$$

$$3\ln 4 + \frac{2}{4} + \frac{3}{16} = 3\ln 4 + 0.5 + 0.1875$$

$$3\ln 1 + \frac{2}{1} + \frac{3}{1^2} = 0 + 2 + 3 = 5$$

$$(3\ln 4 + 0.6875) - 5 = 3\ln 4 - 4.3125$$

$$3\ln 4 - 4.3125$$

Integration by parts

Integration by parts is a very important technique that is used often in engineering and science. It is used to change the integral of a product of functions into an ideally simpler integral. It is the foundation for the theory of differential equations and is used with fourier series.
Is a particularly important technique for integrating a product of two functions.

Introduction to integration by part

From the formula of a derivative of a product:

$$d(u \cdot v) = u' \cdot v + u \cdot v'$$

$$d(u \cdot v) = u' \cdot dx + v \cdot du$$

$$\int u' \cdot v = \int d(u \cdot v) - \int v \cdot du$$

Problem 1: Determine $\int x \cos(x) dx$

$$u = x \rightarrow du = 1 dx = dx$$

$$dx = \cos x \rightarrow v = \sin x$$

$$\int dv = \cos x dx \quad v = \sin x$$

$$\int u' dv = u \cdot v - \int v \cdot du$$

$$\begin{aligned} \int x \cos x dx &= (x)(\sin x) - \int (\sin x) \cdot dx \\ &= x \sin x - \int \sin x dx \\ &= x \sin x - (-\cos x) + C \\ &= x \sin x + \cos x + C \end{aligned}$$

Problem 2: Find $\int 3t e^{(2t)} dt$

$$u = 3t \rightarrow du = 3 dt$$

$$dv = e^{2t} dt \rightarrow v = \frac{1}{2} e^{2t}$$

$$\int dv = \frac{1}{2} \int e^{2t} 2 dt$$

$$a = 2t$$

$$da = 2 dt$$

$$v = \frac{1}{2} e^{2t}$$

$$\int u dv = \int d(uv) - \int v du$$

$$\int 3t e^{2t} dt = (3t) \left(\frac{1}{2} e^{2t} \right) - \int \frac{1}{2} e^{2t} \cdot 3 dt$$

$$= \frac{3}{2} t e^{2t} - \frac{3}{2} \int e^{2t} 2 dt$$

$$a = 2t$$

$$da = 2 dt$$

$$\frac{3}{2} t e^{2t} - \frac{3}{4} e^{2t} + C$$

$$= \frac{3}{2} e^{2t} \left(t - \frac{1}{2} \right) + C$$



Conclusion

The elaboration of this portfolio allowed us to reflect on our learning process during the Integral Calculus course and to recognize the importance of each of the topics studied. Throughout the semester, we consolidated both theoretical and practical knowledge, gaining a better understanding of the usefulness of integrals in mathematical analysis and in applied situations.

The development of exercises and integrative projects represented a challenge that demanded attention to detail, discipline, and structured logical thinking. These aspects strengthened not only our academic formation but also our ability to face problems with greater confidence and method.

In addition, this portfolio constitutes clear evidence of the progress we achieved, showing the transition from the initial comprehension of concepts to the effective application of integration techniques and the resolution of complex problems. We consider that this learning will be highly useful in future courses and in our professional development, as calculus is a fundamental pillar in many areas of knowledge.

In conclusion, this work not only compiles the contents studied in class but also reflects our commitment and effort to achieve meaningful learning, becoming an important foundation for our academic and personal growth.